Aiming for 680–700uF total capacitance
for around 50Hz cut–off.

Note that it does not mean low frequencies
will be perceived significantly lower.
The damping factor below the cut–off is
significantly reduced, which would create
more low end "bloom", and more perceivable
deepness.

More caps are better to distribute current
(make sure ESR is consistent so some caps
are not overworking over others).
This way you would not need big current
ripple rating for the caps.
So you can use smaller values but put
a big grid of them, so the current is
distributed between all of them.

J1
AC power in

With on/off switch
and 3A fuse

±20V / 500W
J4
Switching–Mode Power Supply

R2 0.22 25W
R5 0.22 25W
HS1
DC filter heatsink
C1 10000u 50V
C2 10000u 50V

L1
20 turns for 1mm enameled copper wire over 1cm tube

L2
20 turns for 1mm enameled copper wire over 1cm tube

L3
20 turns for 1mm enameled copper wire over 1cm tube

L4
20 turns for 1mm enameled copper wire over 1cm tube

TR1
Impedance transformer
TR2
Impedance transformer
TR3
Impedance transformer
TR4
Impedance transformer

Tector.it custom toroid
50W RMS, 45Hz–20kHz
0–4–8 Ω BLUE–BROWN–BLACK
0–4–8–16 Ω WHITE–RED–GRAY–GREEN

Each cap ≥1.5 A @100 Hz.
Verify LF frequency derating.

You should expect around 30W of power
for each amplifier channel into 4 ohms.
The idea is that all 4 are driving 4
speakers independently, so that you
get enough power, while each individual
amps is not that powerful.

Everything is designed around 4 ohms
load. The impedance transformers are
supposed to keep that 4 ohms load
consistency regardless of actual
speaker load you connect it to.

Mind that older TDA2050 chips seem to
be rated for lower voltages, so that's
why I picked ±20V supply, because it
seems I have the older model. Otherwise
you could power it from ±24V for more
output power, more headroom. Also be aware
that TDA2050 is obsolete, no longer
manufactured. There are a lot of fakes
in the market.

Sheet: /	
File: wenzels–powered–guitar–cabinet–2609.kicad_sch	
Title: Wenzel's Powered Guitar Cabinet 2609	
Size: A2	Date: September 2026
KiCad E.D.A. 10.0.4	**Rev: r1–wip–1**
	Id: 1/1